Uniform growth of canonical volume on Hilbert modular torsion-level covers over varying totally real fields

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Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

Introduction

Let E be an elliptic curve defined over a number field K. When $K = \mathbb{Q}$, by the celebrated theorem of Mazur [MA78] the torsion part $E(K)_{tor}$ is uniformly bounded. This theorem is generalized to arbitrary number field by Kamienny and Merel:

Theorem. ([Mer96, Kam92]) Let K be a number field of degree d over $\mathbb{Q}$. Then, there exists $N = N(d)$ such that for all elliptic curves E/K we have that

$$E(K)_{tor} \subset E(K)[N].$$

The arithmetic torsion conjecture predicts the same behavior for n-dimensional abelian varieties over number fields:

Conjecture (Arithmetic Torsion Conjecture)**.** For every pair of integers $d, n \geq 1$, there exists an integer $N = N(n, d)$ such that, for every number field K with $[K : \mathbb{Q}] \leq d$ and every n-dimensional abelian variety A over K,

$$A(K)_{\text{tor}} \subseteq A(K)[N].$$

Beyond the case of elliptic curves, this conjecture remains widely open. Its geometric analogue over function fields of curves has been studied in [CT11, CT12, BT18a] and, more recently, in [LY26, GG26]. It is natural to seek an analogue of the geometric torsion conjecture over function fields of higher-dimensional varieties. As explained in [CT11, Appendix A], once the function field of the base is fixed, a torsion bound depending on that function field follows from the curve case by taking suitable hyperplane sections. Our aim is different. We seek a bound that is uniform as the base varies and depends on the base only through an intrinsic birational invariant.

Fix a totally real field $F/\mathbb{Q}$ of degree n and absolute discriminant Δ_F. By an abelian variety with real multiplication by $\mathcal{O}_F$, we mean an n-dimensional abelian variety A equipped with an injection $\mathcal{O}_F \hookrightarrow \mathrm{End}(A)$. We consider such abelian varieties over the function fields of quasi-projective varieties Z of arbitrary dimension and bound the torsion subgroups of their Mordell–Weil groups in terms of the canonical volume of Z. Here, *the canonical volume*

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of Z is the volume of the canonical bundle of any smooth projective variety $\overline{Z}$ birational to Z; that is,

$$\widetilde{\mathrm{vol}}_Z := \mathrm{vol}(K_{\overline{Z}}) = \limsup_{b\to\infty} \frac{h^0(\overline{Z}, bK_{\overline{Z}})}{b^m/m!},$$

where $m = \dim Z$. This quantity is independent of the chosen smooth projective model. In particular, when Z is a curve of geometric genus g, it specializes to $\widetilde{\mathrm{vol}}_Z = \max\{2g-2, 0\}$.

Our first geometric result generalizes [BT18a, Theorem A]:

Theorem A. *(Theorem 4.7) For every real number $v \geq 0$ and positive integer n, there exists an integer $N = N(n, v)$ with the following property. For any totally real field $F/\mathbb{Q}$ of degree n, any complex positive-dimensional integral quasi-projective variety Z with canonical volume at most v, and any abelian scheme $A \to Z$ without isotrivial directions, of relative dimension n and with real multiplication by $\mathcal{O}_F$, the torsion part of the Mordell–Weil group is uniformly bounded:*

$$A(Z)_{\mathrm{tor}} \subseteq A(Z)[N].$$

Explicitly, as explained in Remark 4.8, one may take

$$N(n, v) := \left(\left\lceil (4\pi n \max\{1, v\})^{2n} \right\rceil - 1\right)!.$$

For a fixed totally real field F, there is a Hilbert modular variety $X(1)$ parametrizing abelian varieties with real multiplication by $\mathcal{O}_F$.[1] See Section 1.1 for more details. A family $A \to Z$ of such abelian varieties determines a moduli map

$$\mu_A \colon Z \longrightarrow X(1).$$

We say that $A \to Z$ has *no isotrivial directions* if μ_A is generically finite onto its image. When Z is a curve, this condition is equivalent to non-isotriviality. In higher dimensions, it excludes families pulled back from a lower-dimensional base. This restriction is essential for the canonical volume bound. Indeed, if $B \to C$ is any non-isotrivial family over a curve, then its pullback to $C \times \mathbb{P}^1$ remains non-isotrivial, but $C \times \mathbb{P}^1$ has canonical volume zero.

Remark (**Non-reduction to the curve case**). Theorem A does not follow from the curve case. Indeed, restricting the family to a curve $C \subset Z$, one may obtain a bound in terms of the genus or gonality of C. To recover Theorem A, one would need the resulting lower bounds for these curve invariants to imply a lower bound for the canonical volume of Z. There is no general way to pass from such curvewise bounds to a lower bound for the canonical volume of Z. For example, let Y be a very general abelian variety of dimension $m \geq 4$. Then every curve $C \subset Y$ has gonality at least $\left\lfloor \frac{m}{2} \right\rfloor + 2$ by [Mar20, Corollary 4.7]. Nevertheless, Y has canonical volume zero. This shows that the minimum gonality of curves can be arbitrarily large while the canonical volume of the ambient variety remains zero.

For any ideal $\eta \subset \mathcal{O}_F$, the torsion level cover $X_1(\eta)$ of $X(1)$ parametrizes abelian varieties with real multiplication by $\mathcal{O}_F$, together with a point whose annihilator is η. Accordingly, a torsion section $x \in A(Z)$ with $\mathrm{Ann}_{\mathcal{O}_F}(x) = \eta$ induces a lift of the moduli map

$$\mu_{A,x} \colon Z \longrightarrow X_1(\eta).$$

By a subvariety of $X_1(\eta)$ we mean a positive-dimensional closed, reduced, irreducible subvariety of a smooth projective toroidal compactification $\overline{X_1}(\eta)$ that is not contained in the

[1] For the polarization convention, see Remark 1.1.

boundary. Such a subvariety need not be smooth; it is of general type if a smooth projective model has positive canonical volume. Similar to Theorem A, we obtain the following:

Corollary B. *(Corollary 4.9) Let $F/\mathbb{Q}$ be a totally real field of degree n. Let $X_1(\eta)$ be the cover of the Hilbert modular variety parametrizing abelian varieties with real multiplication by $\mathcal{O}_F$, together with a torsion point having annihilator η.*

Every subvariety of $X_1(\eta)$ is of general type provided that $|\operatorname{Nm}(\eta)| > (2\pi)^{2n}$.

The variety $X_1(\eta)$ is defined over $\mathbb{Q}$, and, given Corollary B, the Bombieri–Lang conjecture predicts that if $|\operatorname{Nm}(\eta)| > (2\pi)^{2n}$, then for every number field K,

$$|X_1(\eta)(K)| < \infty.$$

Although the finiteness predicted by the Bombieri–Lang conjecture remains largely open, combining our effective estimate in Theorem 3.5 with the recent results of Brunebarbe–Maculan [BM24] gives us an unconditional effective bound on the growth rate of K-rational points in the following sense:

Fix a number field K and an ideal η relatively prime to the discriminant Δ_F. Let $\overline{X_1}(\eta)$ be a smooth toroidal compactification of $X_1(\eta)$, with boundary divisor D, and let $X_1(\eta)^*$ be the Baily-Borel compactification of $X_1(\eta)$. Let $L := K_{\overline{X_1}(\eta)} + D$ be the log-canonical bundle on $\overline{X_1}(\eta)$ and let $\pi : \overline{X_1}(\eta) \longrightarrow X_1(\eta)^*$ be the Baily–Borel contraction. Choose $b \geq 1$ and a very ample line bundle A on $X_1(\eta)^*$, defined over $\mathbb{Q}$, such that $\pi^* A_{\mathbb{C}} \simeq bL$. A fixed projective embedding defined by A determines a multiplicative height H_A on $X_1(\eta)^*(\overline{\mathbb{Q}})$. We define

$$H_L(x) := H_A(x)^{1/b}, \qquad x \in X_1(\eta)(\overline{\mathbb{Q}}),$$

and keep this height fixed. See Section 5 for further details.

Theorem C. *(Theorem 5.4) Assume that $|\operatorname{Nm}(\eta)| \geq 5^n$ and that η is relatively prime to the discriminant Δ_F. For every number field K, set $N_{\eta,K}(B) = \#\{x \in X_1(\eta)(K) : H_L(x) \leq B\}$. Then*

$$\limsup_{B\to\infty} \frac{\log \max\{1, N_{\eta,K}(B)\}}{\log B} \leq \delta_{\eta,K,n},$$

where

$$\delta_{\eta,K,n} = \frac{\sqrt{5}[K:\mathbb{Q}]n(n+3)}{|\operatorname{Nm}(\eta)|^{1/(2n)}}.$$

Remark (Sparsity of rational points). In Theorem C, as $|\operatorname{Nm}(\eta)| \to \infty$ through ideals prime to Δ_F, the exponent δ tends to zero, uniformly when $[K:\mathbb{Q}]$ and n are bounded. Therefore, as we go higher in the tower $X_1(\eta)$, its K-rational points become sparser.

Previous results and comparison.

(1) **Geometric torsion over curve bases.** Cadoret–Tamagawa [CT11, CT12] proved a fixed-family version of the geometric torsion conjecture. Their approach uses ℓ-adic monodromy and uniform open-image methods. Bakker and Tsimerman [BT18a] proved the geometric torsion conjecture for abelian varieties with real multiplication, with bounds in terms of the genus of the base curve and, after fixing the field of multiplication, its gonality. Their proof further develops the technique initiated by Hwang and To in [HT02, HT20, HT00]. More recently, Looper and Yap [LY26] proved the geometric torsion conjecture in characteristic zero for arbitrary traceless abelian varieties, with a

bound depending on the gonality of the base curve. Their approach follows the strategy of Hindry–Silverman. Gao and Gu [GG26] subsequently gave a different proof, inspired by Vojta's method, and obtained explicit genus-dependent bounds.

Our approach is closer in spirit to that of Bakker–Tsimerman. We establish quantitative hyperbolicity properties of torsion level covers of Hilbert modular varieties by obtaining lower bounds for the canonical volumes of their subvarieties. These hyperbolicity results also show that the Bombieri–Lang conjecture predicts strong finiteness properties for rational points on sufficiently deep torsion level covers. Since our estimates apply to subvarieties of arbitrary dimension, they also yield quantitative sparsity results for rational points on these covers. Such sparsity results seem not to be within reach of methods that study only curves in the moduli space.

(2) **Full level versus torsion level structures.** Hyperbolicity in towers of full-level congruence covers of locally symmetric varieties has been studied extensively; see, for example, [Nad89, HT06, Bru20a, Bru20b, AVA18, AVA17, WY23, WY25], and [KL22] in positive characteristic. Brunebarbe's results in particular give canonical volume growth for subvarieties in the towers of covering with full level structure. A crucial mechanism in the full level structure is that the covering maps ramify to increasingly high order along every boundary component. This ramification allows one to use the positivity gained along the tower to deduce hyperbolicity.

The torsion level covers considered here behave differently. The space $X_1(\eta)$ parametrizes a single torsion point rather than a full level structure, and the map $X_1(\eta) \to X(1)$ does not have uniformly increasing ramification at all cusps. This phenomenon already occurs for modular curves: the map $X_1(p) \to X(1)$ has $(p-1)/2$ cusps that are unramified; see [Shi71, p. 26] and [Ogg73]. Thus, the standard argument for the full level covers cannot be applied directly to the torsion level covers.

(3) **Rational points versus integral points.** It is observed in [ELV23, BM24] that if one has control over the degree of all subvarieties, the bound on the growth rate of rational points improves in the strategy of Bombieri-Pila [BP89] and Heath-Brown [HB02]. However, to get the lower bound on the degree of subvarieties, they passed to étale covers and this restricts them to obtain a result only on the integral points, rather than rational points. The point is that for rational points on a nonproper variety, the associated fiber torsors need not be unramified outside a common finite set of places, so one cannot in general reduce to finitely many twists. For integral points, by contrast, such uniform ramification control is available (see Remark 5.6 for a precise discussion).

Our intrinsic approach has the advantage that it does not require passage to a cover to raise the degree of subvarieties and hence we can get the bound on the growth rate of rational points. In general, bounding the growth rate of rational points on a quasi-projective variety is more difficult than bounding the growth rate of integral points. For example, on $X = \mathbb{P}^1_K \setminus \{0, 1, \infty\}$, there are infinitely many K-rational points, however, there are only finitely many integral points on X because of the famous theorem of Siegel and the growth rate of integral points in this case is zero (see [BM24, Remark 3.3]).

(4) **Kodaira dimension of Hilbert modular varieties.** Tsuyumine [Tsu85] proved that Hilbert modular varieties (without level structure) are of general type when $n > 6$, by constructing pluricanonical forms from Hilbert modular forms and controlling their vanishing along the cusps and exceptional divisors. His estimates were later used in

[CRT19, Theorem D] to show, up to finitely many exceptions, all subvarieties are of general type outside a proper exceptional locus. This approach, however, does not appear to identify that exceptional locus.

Bakker–Tsimerman [BT18a, Corollary G] recovered Tsuyumine's result by a different method. Building on their work, we show that, after passing to an effectively computable sufficiently deep level, every subvariety not contained in the boundary divisor D is of general type. Thus, unlike the exceptional locus in [CRT19, Theorem D], the exceptional locus in our result is explicitly controlled: it is contained in the boundary divisor D.

Strategy of proofs. Theorem A: The core idea is to identify a region in the locally symmetric space that is both "algebraically thick" and "geometrically thick": every subvariety must intersect this region, and its injectivity radius has a lower bound tending to infinity on the tower of torsion covers. Once such a region is found, we apply a theorem of Hwang–To to obtain lower bounds for the hyperbolic volumes of subvarieties. The passage from hyperbolic volume to canonical volume is provided by the higher-dimensional Arakelov inequality recently proved by Brunebarbe [Bru20a, Theorem 1.6]. To apply this inequality, we must subtract a fixed multiple of the boundary divisor from the log-canonical bundle. This is the only step for which we use the earlier result of Bakker–Tsimerman.

In this paper, the thick part is the complement of the canonical horoball neighborhoods of the cusps. In Theorem 3.1, we prove, using the monodromy of subvarieties, that every subvariety of $X_1(\eta)$ meets this thick part. In Proposition 3.4, we show that, provided $X_1(\eta)$ has no elliptic points, the injectivity radius of its thick part has a lower bound depending only on $[F:\mathbb{Q}]$ and $|\mathrm{Nm}(\eta)|$. This estimate is uniform in F; it depends on the field of multiplication only through its degree n, and is independent of the choice of toroidal compactification.

Compared with the full-level results mentioned above, our "increasing hyperbolicity" results arise from the growth of the injectivity radius on the thick part, rather than from ramification at the cusps. Compared with the ball quotient case in [Mem25], the thick region for Hilbert modular varieties is defined differently. The key properties required of the thick region are the same, but its definition and the arguments used to establish these properties are somewhat different.

Theorem C: We combine the estimate for log-canonical degrees from Theorem 3.5 with Brunebarbe–Maculan's recent refinement of the determinant method in [BM24, Theorem 4.4].

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1. Hilbert modular varieties

We recall the setup for Hilbert modular varieties, following [BT18a, §§1 and 3]. In particular, we adopt the conventions of [BT18a, §1] regarding cusps, their neighborhoods, and toroidal compactifications. We use the notation and definitions introduced here throughout

the paper without further reference. All estimates below are uniform in the choice of the projective $\mathcal{O}_F$-module M.

1.1. **Hilbert modular varieties.** Let $\mathbb{H} = \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}$ denote the upper half-plane, and let

$$\mathbb{H}^n := \underbrace{\mathbb{H} \times \cdots \times \mathbb{H}}_{n \text{ times}}$$

be the product of n copies of the upper half plane. The group $G = \mathrm{SL}_2(\mathbb{R})^n$ acts holomorphically on $\mathbb{H}^n$ by Möbius transformations component-wise.

Let F be a totally real field of degree n, with ring of integers $\mathcal{O}_F$ and real embeddings $\sigma_j : F \hookrightarrow \mathbb{R}$, $j = 1, \ldots, n$; write $\mathfrak{d}_F$ for its different and $\Delta_F = |\operatorname{Nm}(\mathfrak{d}_F)|$ for its absolute discriminant. Fix a projective $\mathcal{O}_F$-module M of rank 2. Since $\mathcal{O}_F$ is a Dedekind domain, there is a nonzero fractional ideal $\mathfrak{a}$ such that $M \cong \mathcal{O}_F \oplus \mathfrak{a}$. Replacing $\mathfrak{a}$ by an integral ideal in the same ideal class, we may assume that $\mathfrak{a} \subset \mathcal{O}_F$. The Hilbert modular group associated with M is

$$\Gamma(1) := \mathrm{SL}(M) = \{g \in \mathrm{Aut}_{\mathcal{O}_F}(M) \mid \det_F(g) = 1\},$$

where $\det_F(g)$ denotes the determinant of the induced F-linear automorphism of $M \otimes_{\mathcal{O}_F} F$. Relative to $M = \mathcal{O}_F \oplus \mathfrak{a}$, its elements have the form

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \alpha, \delta \in \mathcal{O}_F, \quad \beta \in \mathfrak{a}^{-1}, \quad \gamma \in \mathfrak{a}, \quad \alpha\delta - \beta\gamma = 1.$$

For each j, the identification $M_j := M \otimes_{\mathcal{O}_F, \sigma_j} \mathbb{R} \cong \mathbb{R}^2$ sends g to

$$g_j = \begin{pmatrix} \sigma_j(\alpha) & \sigma_j(\beta) \\ \sigma_j(\gamma) & \sigma_j(\delta) \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R}).$$

The resulting homomorphism

$$\Gamma(1) \longrightarrow \mathrm{SL}_2(\mathbb{R})^n, \qquad g \longmapsto (g_1, \ldots, g_n),$$

is injective and has arithmetic image. We henceforth identify $\Gamma(1)$ with this image. It acts on $\mathbb{H}^n$ by

$$g \cdot (z_1, \ldots, z_n) = \left(\frac{\sigma_j(\alpha) z_j + \sigma_j(\beta)}{\sigma_j(\gamma) z_j + \sigma_j(\delta)} \right)_{j=1}^{n},$$

and we set

$$X(1) := \Gamma(1) \backslash \mathbb{H}^n.$$

We regard $X(1)$ with its natural orbifold structure, while its underlying coarse analytic space is the usual Hilbert modular variety. More generally, for a finite-index subgroup $\Gamma \subset \Gamma(1)$, we write

$$X_\Gamma := \Gamma \backslash \mathbb{H}^n.$$

Since Γ is an arithmetic lattice, the quotient X_Γ is a quasi-projective variety. Its cusps are the Γ-equivalence classes in $\mathbb{P}^1(F)$, of which there are only finitely many. By the Baily–Borel theorem [BB66], X_Γ admits a normal projective compactification X_Γ^* obtained by adjoining one point for each cusp. If Γ is torsion-free, then X_Γ is a smooth complex variety and $X_\Gamma \to X(1)$ is a finite orbifold cover.

Over $\mathbb{C}$, the Hilbert modular stack $X(1)$ associated with $\Gamma(1) = \mathrm{SL}(M)$ is naturally identified with the moduli stack of abelian varieties A of dimension n, equipped with real multiplication $\iota : \mathcal{O}_F \hookrightarrow \mathrm{End}(A)$, such that

$$H_1(A, \mathbb{Z}) \cong M$$

as $\mathcal{O}_F$-modules, where the $\mathcal{O}_F$-module structure on $H_1(A, \mathbb{Z})$ is induced by ι.

Remark 1.1. Strictly speaking, the moduli problem fixes a polarized projective $\mathcal{O}_F$-module type (M, ψ) and parametrizes triples (A, ι, λ), where λ is a polarization satisfying $\lambda \circ \iota(a) = \iota(a)^\vee \circ \lambda$ for every $a \in \mathcal{O}_F$, and the displayed marking preserves the polarized type. For a family with real multiplication over an integral base, one may choose λ on the generic fiber and extend it after shrinking the base; its type is then constant [Gor02, Lemma 2.12 and Theorem 2.16]. Only this fixed choice is used. Following [BT18a, § 3], we suppress λ and ψ from the notation; all estimates are uniform in this choice.

For a nonzero ideal $\eta \subset \mathcal{O}_F$, let

$$\Gamma_1(\eta) := \left\{ \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \Gamma(1) \;\middle|\; \alpha \equiv 1 \pmod{\eta}, \quad \gamma \in \eta\mathfrak{a} \right\}.$$

This is precisely the stabilizer of the class of $(1, 0)$ in $M/\eta M$. We denote the corresponding level cover by

$$X_1(\eta) := \Gamma_1(\eta)\backslash\mathbb{H}^n.$$

For an abelian variety (A, ι) with real multiplication, set

$$A[\eta] := \bigcap_{\alpha \in \eta} \ker\big(\iota(\alpha) : A \longrightarrow A\big).$$

With the convention of Remark 1.1, $X_1(\eta)$ parametrizes tuples (A, ι, P), where $H_1(A, \mathbb{Z}) \cong M$ as $\mathcal{O}_F$-modules and

$$P \in A[\eta](\mathbb{C}), \qquad \mathrm{Ann}_{\mathcal{O}_F}(P) = \eta.$$

Lemma 1.2. *If $|\mathrm{Nm}(\eta)| > 4^n$, then $\Gamma_1(\eta)$ acts freely on $\mathbb{H}^n$, and every parabolic element of $\Gamma_1(\eta)$ is unipotent.*

Proof. By [BT18a, Lemma 3.1], $\Gamma_1(\eta)$ acts freely on $\mathbb{H}^n$. Let $g = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \Gamma_1(\eta)$ be parabolic. Since $\alpha \equiv 1 \pmod{\eta}$ and $\beta\gamma \in \eta$, the determinant relation gives $\delta \equiv 1 \pmod{\eta}$. Therefore, $\mathrm{tr}(g) \equiv 2 \pmod{\eta}$. Since g is parabolic, its trace is 2 or -2. The second case would imply $4 \in \eta$, and hence $|\mathrm{Nm}(\eta)| \leq 4^n$, which is a contradiction. It follows that $\mathrm{tr}(g) = 2$, so g is unipotent. □

1.2. **Neighborhoods of cusps and a toroidal compactification.** The boundary of each factor $\mathbb{H}$ is identified with $\partial\mathbb{H} = \mathbb{P}^1(\mathbb{R})$. Möbius transformations in $\mathrm{SL}_2(\mathbb{R})$ act on this boundary as well. Applying this in every factor extends the component-wise action of $\Gamma \subset \mathrm{SL}_2(\mathbb{R})^n$ from $\mathbb{H}^n$ to $\mathbb{P}^1(\mathbb{R})^n$. The real embeddings of F define an embedding

$$\mathbb{P}^1(F) \hookrightarrow \mathbb{P}^1(\mathbb{R})^n, \qquad [a : b] \longmapsto \big([\sigma_1(a) : \sigma_1(b)], \ldots, [\sigma_n(a) : \sigma_n(b)]\big).$$

For the Hilbert modular groups considered here, the rational boundary components are precisely the points arising from $\mathbb{P}^1(F)$ under this embedding. A cusp of X_Γ is a Γ-equivalence class of such rational boundary points, and there are only finitely many such equivalence classes.

Let c be a cusp and choose a representative $[a:b] \in \mathbb{P}^1(F)$. Since $\mathrm{SL}_2(F)$ acts transitively on $\mathbb{P}^1(F)$, there exists $g \in \mathrm{SL}_2(F)$ such that $g \cdot [a:b] = [1:0]$, which corresponds to $\infty := (\infty, \dots, \infty) \in \mathbb{P}^1(\mathbb{R})^n$. Replacing Γ by the conjugate lattice $g\Gamma g^{-1}$ gives an isomorphism

$$\Gamma\backslash\mathbb{H}^n \xrightarrow{\sim} (g\Gamma g^{-1})\backslash\mathbb{H}^n, \qquad [z] \longmapsto [g \cdot z].$$

After conjugating Γ, we may assume that the chosen cusp c is represented by ∞. This choice is unique up to left multiplication by an element of the stabilizer of ∞, namely the upper-triangular subgroup

$$P_\infty(\mathbb{R}) = \left\{ p(a,b) : a \in (\mathbb{R}^\times)^n,\ b \in \mathbb{R}^n \right\} \subset \mathrm{SL}_2(\mathbb{R})^n, \tag{1.1}$$

where, for $a = (a_1, \dots, a_n) \in (\mathbb{R}^\times)^n$, and $b = (b_1, \dots, b_n) \in \mathbb{R}^n$, we set

$$p(a,b) = \left(\begin{pmatrix} a_1 & b_1 \\ 0 & a_1^{-1} \end{pmatrix}, \dots, \begin{pmatrix} a_n & b_n \\ 0 & a_n^{-1} \end{pmatrix} \right).$$

Let

$$N(z) := \prod_{j=1}^n \mathrm{Im}(z_j), \qquad U(s) := \{ z \in \mathbb{H}^n : N(z) > 1/s \}.$$

We call $U(s)$ the horoball of depth s based at ∞. Since $N\big(p(a,b)z\big) = \left(\prod_{j=1}^n a_j^2\right) N(z)$, we have $p(a,b)U(s) = U(s')$, where $s' = s/\prod_{j=1}^n a_j^2$.

Let $\Gamma_\infty = \Gamma \cap P_\infty(\mathbb{R})$ be the stabilizer of ∞, and let $\Lambda_\infty \lhd \Gamma_\infty$ be its subgroup of unipotent translations, identified with a lattice in $\mathbb{R}^n$. Using $\Lambda_\infty \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}^n$, we identify $\mathbb{H}^n$ with the positive cone

$$\mathcal{H}_\infty := \{ \zeta \in \Lambda_\infty \otimes_{\mathbb{Z}} \mathbb{C} \mid (\mathrm{Im}\,\zeta)_j > 0 \text{ for all } j \}.$$

The lattice Λ_∞ acts on $\mathcal{H}_\infty$ by translations $\Lambda_\infty \otimes 1$. The function N induces the product norm

$$\mathrm{Nm}_\infty(\lambda) := \prod_{j=1}^n \lambda_j, \qquad \lambda = (\lambda_1, \dots, \lambda_n) \in \Lambda_\infty.$$

Under conjugation by $p(a,b)$, both N and Nm_∞ are multiplied by $\prod_{j=1}^n a_j^2$. Similar to [BT18a, §1], we normalize them so that

$$\min_{0 \neq \lambda \in \Lambda_\infty} |\mathrm{Nm}_\infty(\lambda)| = 1. \tag{1.2}$$

This normalization removes the ambiguity coming from the choice of coordinates at the cusp. More generally, let c_i be a cusp with stabilizer Γ_i and translation lattice Λ_i. The same normalization gives coordinate forms $\sigma_i^j : \Lambda_i \longrightarrow \mathbb{R}$ and a function $N_i(\zeta) := \prod_{j=1}^n \sigma_i^j\big(\mathrm{Im}\,\zeta\big)$ on the corresponding positive cone. Although the individual coordinates σ_i^j depend on the choice of coordinates at the cusp, their normalized product N_i does not. For $s > 0$, let

$$U_i(s) := \{ \zeta \in \mathcal{H}_i \mid N_i(\zeta) > 1/s \},$$

and let $\iota_{i,s} : \Gamma_i\backslash U_i(s) \to X_\Gamma$ be the natural map.

Definition 1.3. ([BT18a, Def 1.1]) The *uniform depth* d of the cusps of X_Γ is the largest $s > 0$ such that

(1) the map $\iota_{i,s}$ is injective for every cusp c_i;

(2) the subsets $\iota_{i,s}\big(\Gamma_i\backslash U_i(s)\big) \subset X_\Gamma$ are pairwise disjoint[2].

For $0 < s \le d$, we identify $\Gamma_i\backslash U_i(s)$ with its image in X_Γ and write

$$H_i(s) := \iota_{i,s}\big(\Gamma_i\backslash U_i(s)\big) \subset X_\Gamma.$$

Toroidal compactifications of X_Γ were constructed in [AMRT10]. Their construction requires the choice, at each cusp c_i, of a rational polyhedral fan in the corresponding positive cone that is admissible with respect to the action of Γ_i/Λ_i. The resulting compactification depends on these choices, although any two choices admit a common refinement. When Γ has no elliptic points, we choose a regular projective admissible collection of fans. The associated toroidal compactification $\overline{X}_\Gamma$ is then a smooth projective variety, and the reduced boundary divisor $D := \big(\overline{X}_\Gamma \setminus X_\Gamma\big)_{\mathrm{red}}$ is a simple normal crossings divisor; see [BT18a, §1, §4] for more details. We fix such a choice throughout.

Definition 1.4. The *canonical horoball* of the cusp c_i is

$$H_i := H_i(d) \subset X_\Gamma \subset \overline{X}_\Gamma,$$

where d is the uniform depth of the cusps of Γ.

Similar to [Mem25, §4], we isolate a compact part of X_Γ by removing all its canonical horoballs:

Definition 1.5. We define the *thick part* of X_Γ to be

$$X_{\Gamma,\mathrm{thick}} := X_\Gamma \setminus \bigcup_i H_i = \overline{X}_\Gamma \setminus \left(D \cup \bigcup_i H_i \right).$$

Following [BT18a, §1, pp. 7–8], we recall the local coordinates near the boundary D. For a cusp c_i, set $T_i := \Lambda_i\backslash\big(\Lambda_i \otimes_{\mathbb{Z}} \mathbb{C}\big)$. This is an algebraic torus with character lattice $\Lambda_i^\vee := \mathrm{Hom}_{\mathbb{Z}}(\Lambda_i, \mathbb{Z})$. For $\chi \in \Lambda_i^\vee$, the corresponding character is

$$q^\chi(\zeta) := \exp\big(2\pi\sqrt{-1}\,\chi(\zeta)\big).$$

Indeed, this function is invariant under $\zeta \mapsto \zeta + \lambda$ for $\lambda \in \Lambda_i$, since $\chi(\lambda) \in \mathbb{Z}$. Let τ be a maximal cone in the chosen regular fan. Its primitive ray generators form a basis of Λ_i; if $\chi_1, \dots, \chi_n$ is the dual basis, then the associated toric chart has coordinates $q_k := q^{\chi_k}$ and the toric boundary is given by $q_1 \cdots q_n = 0$.

1.3. **Metric and hyperbolic volume.** Endow $\mathbb{H}$ with its hyperbolic hermitian metric $ds^2_{\mathbb{H}}$ of constant sectional curvature -1; explicitly, the associated Kähler form is

$$\omega_{\mathbb{H}} = \frac{1}{2}\,\mathrm{Im}\, ds^2_{\mathbb{H}} = \frac{i dz \wedge d\bar{z}}{2y^2} = i\partial\bar{\partial}(-2\log y)$$

Likewise endow $\mathbb{H}^n$ with the invariant metric $ds^2_{\mathbb{H}^n} = \sum_i \pi_i^* ds^2_{\mathbb{H}}$, where $\pi_j : \mathbb{H}^n \to \mathbb{H}$ is the jth projection, and we denote by $\omega_{\mathbb{H}^n}$ the associated Kähler form. We use this Kähler metric to define hyperbolic volume. For the injectivity-radius estimates, we use the Kobayashi distance on $\mathbb{H}^n$:

$$d_{\mathbb{H}^n}(z, w) := \max_{1\le j\le n} d_{\mathbb{H}}(z_j, w_j).$$

[2]Pairwise disjointness is not mentioned in [BT18a, Definition 1.1], but it is verified separately for the depths considered in [BT18a, §3] and used throughout the paper. We incorporate this condition into the definition, as in [BT18b, Definition 3.7].

Let $\Gamma \subset G$ be a discrete non-degenerate subgroup of finite covolume, and let $X = \Gamma\backslash\mathbb{H}^n$ be endowed with the induced Kähler metric ds^2_X and Kähler form ω_X. Let $\overline{X}$ be a smooth toroidal compactification of X, with boundary divisor $D = \overline{X}\backslash X$. By [Mum77, Theorem 3.1, Proposition 3.4(a)], the invariant metric on Ω^1_X is a good metric on $\Omega^1_{\overline{X}}(\log D)$. Hence its determinant metric h is good on $\det \Omega^1_{\overline{X}}(\log D) = K_{\overline{X}} + D$. Since the first Chern form of h is $\omega_X/(2\pi)$, [Mum77, Propositions 1.1–1.2, Theorem 1.4] shows that the trivial extension of $\omega_X/(2\pi)$ is a closed current, with no boundary residue, representing

$$c_1(K_{\overline{X}} + D) = \frac{1}{2\pi}[\omega_X] \in H^{1,1}(\overline{X}, \mathbb{R}). \tag{1.3}$$

Throughout the paper, all varieties and subvarieties are assumed to be irreducible and of positive dimension. Let $Z \subset \overline{X}$ be an m-dimensional subvariety not contained in D, and set $Z^\circ := Z \cap X$. The *hyperbolic volume* of Z is

$$\mathrm{vol}_X(Z) := \frac{1}{m!}\int_{(Z^\circ)_{\mathrm{reg}}} \omega_X^m,$$

where $(Z^\circ)_{\mathrm{reg}}$ is the smooth locus of Z°. By Mumford's theorem for good metrics [Mum77], applied after taking a log resolution of $(Z, D_{|Z})$, this integral is finite and

$$(K_{\overline{X}} + D)^m \cdot Z = \frac{1}{(2\pi)^m}\int_{(Z^\circ)_{\mathrm{reg}}} \omega_X^m = \frac{m!}{(2\pi)^m}\,\mathrm{vol}_X(Z). \tag{1.4}$$

We recall the Kobayashi-ball form of the volume estimate of Hwang–To. Here volume is computed with respect to ω_X, while the injectivity radius is computed with respect to the Kobayashi distance defined above.

For $X = \Gamma\backslash\mathbb{H}^n$, with torsion-free lattice Γ, the *injectivity radius* at $x \in X$ is

$$\mathrm{inj}_X(x) := \frac{1}{2}\inf_{\gamma\in\Gamma\backslash\{1\}} d_{\mathbb{H}^n}(z, \gamma z),$$

where $z \in \mathbb{H}^n$ is any lift of x. We recall the special case of result of Hwang and To on bounding the hyperbolic volume from below:

Theorem 1.6. *([HT02, Theorem 1.1]) Let Z be an m-dimensional subvariety of X passing through $x \in X$. Let $r = \mathrm{inj}_x(X)$ be the injectivity radius of x on X. Then, the following inequality holds:*

$$\mathrm{vol_X(Z)} \geq \frac{(4\pi)^m}{m!}\sinh^{2m}(r/2)\cdot \mathrm{mult}_x(Z), \tag{1.5}$$

where $\mathrm{mult}_x(Z)$ is the multiplicity of x on Z.

With some modifications, Theorem 1.6 can also be stated without assuming that Γ is torsion-free; see [Li25, Theorem 4.1] for the case of two-dimensional ball quotients.

2. Hodge-Theoretic preliminaries

We recall the Hodge-theoretic facts needed for the proofs of Theorems 3.1 and Proposition 4.2. In particular, we prove Proposition 2.1, which is used in the proof of Proposition 4.2. Our conventions for the Griffiths line bundle and its extension follow [Bru20a, §§2–3].

2.1. **The standard variation of Hodge structure.** Let $\Gamma \subset \Gamma(1)$ be a torsion-free finite-index subgroup. We will recall the standard integral variation of Hodge structure on X_Γ. Set $M_\mathbb{Q} := M \otimes_\mathbb{Z} \mathbb{Q}$, $V_\mathbb{Z} := \mathrm{Hom}_\mathbb{Z}(M, \mathbb{Z})$, and $V_\mathbb{Q} := V_\mathbb{Z} \otimes_\mathbb{Z} \mathbb{Q} = \mathrm{Hom}_\mathbb{Q}(M_\mathbb{Q}, \mathbb{Q})$. Fiberwise, the identification $H_1(A, \mathbb{Z}) \cong M$ identifies $V_\mathbb{Z}$ with $H^1(A, \mathbb{Z})$. The action of Γ on $M_\mathbb{Q}$ induces the monodromy representation

$$\rho : \Gamma \longrightarrow \mathrm{GL}(V_\mathbb{Q}), \qquad \rho(\gamma)(\ell) := \ell \circ \gamma^{-1}.$$

The diagonal action $\gamma \cdot (z, \ell) := (\gamma z, \rho(\gamma)\ell)$ therefore defines an integral local system

$$\mathbb{V}_\mathbb{Z} := \Gamma \backslash \big(\mathbb{H}^n \times V_\mathbb{Z}\big)$$

on X_Γ. Since M has rank 2 over $\mathcal{O}_F$, the local system $\mathbb{V}_\mathbb{Z}$ has rank $2n$. We write $\mathbb{V}_\mathbb{Q} := \mathbb{V}_\mathbb{Z} \otimes_\mathbb{Z} \mathbb{Q}$, and $\mathbb{V}_\mathbb{C} := \mathbb{V}_\mathbb{Z} \otimes_\mathbb{Z} \mathbb{C}$. The local system $\mathbb{V}_\mathbb{C}$ underlies the holomorphic bundle

$$\mathcal{V} := \mathbb{V}_\mathbb{C} \otimes_\mathbb{C} \mathcal{O}_{X_\Gamma}$$

endowed with its tautological flat connection ∇, whose monodromy is ρ. The corresponding polarization on the standard cohomological local system induces, with the sign chosen so that it polarizes H^1, a Γ-invariant nondegenerate alternating form Q_V on $V_\mathbb{Q}$. After multiplying Q_V by a positive integer, we may assume that its restriction to $V_\mathbb{Z}$ is integral, and it therefore gives a flat pairing on $\mathbb{V}_\mathbb{Z}$. The real embeddings of F give a decomposition

$$V_\mathbb{R} := V_\mathbb{Z} \otimes_\mathbb{Z} \mathbb{R} = \bigoplus_{j=1}^{n} V_j, \qquad V_j := \big(M \otimes_{\mathcal{O}_F, \sigma_j} \mathbb{R}\big)^\vee.$$

Since the polarization is compatible with the $\mathcal{O}_F$-action, the summands V_j are pairwise Q_V-orthogonal. Starting from the basis of V_j dual to the standard basis of $M \otimes_{\mathcal{O}_F, \sigma_j} \mathbb{R} \cong \mathbb{R}^2$, we rescale both vectors by the same positive real number and denote the resulting basis by (e_j, f_j), so that $Q_V(e_j, f_j) = -1$. For $z = (z_1, \ldots, z_n) \in \mathbb{H}^n$, define

$$F^1_z := \bigoplus_{j=1}^{n} \mathbb{C}(e_j - z_j f_j) \subset V_\mathbb{C}.$$

If $\gamma_j = \begin{pmatrix} a_j & b_j \\ c_j & d_j \end{pmatrix}$ is the image of $\gamma \in \Gamma$ in the j-th factor, then

$$\gamma_j^{-T} \begin{pmatrix} 1 \\ -z_j \end{pmatrix} = (c_j z_j + d_j) \begin{pmatrix} 1 \\ -\gamma_j z_j \end{pmatrix}. \tag{2.1}$$

Hence $\rho(\gamma) F^1_z = F^1_{\gamma z}$. The spaces F^1_z therefore descend to a holomorphic subbundle

$$F^1\mathcal{V} \subset \mathcal{V}.$$

Together with $F^0\mathcal{V} = \mathcal{V}$ and $F^2\mathcal{V} = 0$, this defines a weight-one Hodge filtration. Since $z_j \notin \mathbb{R}$, one has $V_\mathbb{C} = F^1_z \oplus \overline{F^1_z}$, and hence $h^{1,0} = h^{0,1} = n$. Moreover, $Q_V(F^1_z, F^1_z) = 0$, and for $v = \sum_{j=1}^n \alpha_j (e_j - z_j f_j) \in F^1_z$ one has $\sqrt{-1}\, Q_V(v, \overline{v}) = 2 \sum_{j=1}^n \mathrm{Im}(z_j) |\alpha_j|^2 > 0$ whenever $v \neq 0$. Hence $(\mathbb{V}_\mathbb{Z}, F^\bullet\mathcal{V}, Q_V)$ is a polarized integral variation of Hodge structure of weight 1 on X_Γ and the complex variation $\mathbb{V}_\mathbb{C}$ has length 1.

2.2. **The Hodge bundle.** Let $\Gamma := \Gamma_1(\eta)$ and $X := X_\Gamma = X_1(\eta)$. We assume that $|\operatorname{Nm}(\eta)| > 4^n$ so that Γ is torsion free and all parabolic elements are unipotent by Lemma 1.2. Let $\pi : \mathcal{A} \to X$ be the universal abelian scheme. Its Hodge bundle is

$$\mathbb{E} := \pi_* \Omega^1_{\mathcal{A}/X} = F^1 \mathcal{V}.$$

For a point $x \in X$, corresponding to an abelian variety A_x, the fiber of $\mathbb{E}$ is $\mathbb{E}_x = H^0(A_x, \Omega^1_{A_x})$. Note that by the real multiplication by $\mathcal{O}_F$, we have an embedding $\iota : \mathcal{O}_F \hookrightarrow \operatorname{End}_X(\mathcal{A})$. Therefore, every element $a \in \mathcal{O}_F$ gives an endomorphism

$$\iota(a) : \mathcal{A} \longrightarrow \mathcal{A}.$$

The action of $\mathcal{O}_F$ on $\mathcal{A}$ induces an action of $\mathcal{O}_F$ on the Hodge bundle by pullback. Namely, for each $a \in \mathcal{O}_F$, we obtain an $\mathcal{O}_X$-linear endomorphism $\iota(a)^* : \mathbb{E} \longrightarrow \mathbb{E}$. Therefore $\mathbb{E}$ is naturally an $(\mathcal{O}_F \otimes_{\mathbb{Z}} \mathcal{O}_X)$-module. After base change to $\mathbb{C}$, the algebra $\mathcal{O}_F$ splits via the real embeddings $\sigma_1, \dots, \sigma_n : F \hookrightarrow \mathbb{R} \subset \mathbb{C}$:

$$\mathcal{O}_F \otimes_{\mathbb{Z}} \mathbb{C} \cong \bigoplus_{j=1}^{n} \mathbb{C}, \qquad a \otimes z \longmapsto \big(z\sigma_1(a), \dots, z\sigma_n(a)\big).$$

Hence, over X, we have $\mathcal{O}_F \otimes_{\mathbb{Z}} \mathcal{O}_X \cong \bigoplus_{j=1}^n \mathcal{O}_X$. This decomposition gives idempotent projectors $p_1, \dots, p_n$, where p_j is the projector onto the j-th factor. These satisfy

$$p_j^2 = p_j, \qquad p_i p_j = 0 \quad \text{for } i \neq j, \qquad p_1 + \cdots + p_n = 1.$$

Since these projectors act on $\mathbb{E}$, they split the Hodge bundle into their images:

$$\mathbb{E} = \bigoplus_{j=1}^{n} \mathcal{L}_j,$$

where $\mathcal{L}_j := p_j \mathbb{E}$ (see for example [Gor02, §2.3]). Each $\mathcal{L}_j$ is the automorphic line bundle corresponding to the j-th embedding. Fiberwise, for a point corresponding to an abelian variety A, one has

$$(\mathcal{L}_j)_A = \left\{ \omega \in H^0(A, \Omega^1_A) \mid a^*\omega = \sigma_j(a)\omega \text{ for all } a \in \mathcal{O}_F \right\}.$$

The pull back of $\mathcal{L}_j$ on $\mathbb{H}^n$ is generated by the section $s_j(z) := e_j - z_j f_j$. For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$, we write $j_j(\gamma, z) := \sigma_j(c) z_j + \sigma_j(d)$. Then (2.1) gives

$$\rho(\gamma) s_j(z) = j_j(\gamma, z) s_j(\gamma z).$$

Therefore, $\mathcal{L}_j$ is the automorphic line bundle of weight one at the embedding σ_j.

Let $\lambda := \det \mathbb{E}$. In additive notation in Picard group, this becomes $\lambda = \mathcal{L}_1 + \cdots + \mathcal{L}_n$. Let $j : X \hookrightarrow \overline{X}$ be a smooth toroidal compactification with simple normal-crossings boundary $D = \overline{X} \setminus X$. A small loop around an irreducible component of D is represented by an element of the unipotent translation lattice of the corresponding cusp. Its action on the standard representation, and hence on $\mathbb{V}_{\mathbb{C}}$, is therefore unipotent.

The flat bundle $(\mathcal{V}, \nabla)$ introduced above admits its canonical Deligne extension $(\overline{\mathcal{V}}, \overline{\nabla})$ to $\overline{X}$, characterized by the fact that

$$\overline{\nabla} : \overline{\mathcal{V}} \longrightarrow \overline{\mathcal{V}} \otimes \Omega^1_{\overline{X}}(\log D)$$

has nilpotent residues along the irreducible components of D. The Hodge filtration extends by setting

$$\overline{F}^p := \overline{\mathcal{V}} \cap j_*(F^p\mathcal{V}) \subset j_*\mathcal{V}.$$

These are locally free subbundles of $\overline{\mathcal{V}}$; the resulting filtered logarithmic bundle $(\overline{\mathcal{V}}, \overline{\nabla}, \overline{F}^\bullet)$ is the Deligne–Schmid extension of the variation. Since the local monodromies are unipotent, the associated parabolic structure has only weight zero. Thus, the parabolic extension is the Deligne–Schmid extension endowed with the trivial parabolic structure; see [Bru20a, §1.2, §2.2]. Let

$$\overline{\mathbb{E}} := \overline{F}^1, \qquad \overline{\lambda} := \det \overline{\mathbb{E}}.$$

Every element of $\mathcal{O}_F$ acts on $\mathcal{V}$ by a flat endomorphism preserving $F^1\mathcal{V}$. By the functoriality of the canonical Deligne extension, this action extends to $\overline{\mathcal{V}}$ and preserves $\overline{\mathbb{E}}$. The idempotent projectors $p_1, \dots, p_n$ therefore extend to $\overline{\mathbb{E}}$, and yield a decomposition

$$\overline{\mathbb{E}} = \bigoplus_{j=1}^n \overline{\mathcal{L}}_j, \qquad \overline{\mathcal{L}}_j := p_j\overline{\mathbb{E}}.$$

Here $\overline{\mathcal{L}}_j$ is the eigenline on which $a \in \mathcal{O}_F$ acts through $\sigma_j(a)$, and $\overline{\mathcal{L}}_j|_X \simeq \mathcal{L}_j$. In particular,

$$\overline{\lambda} = \det \overline{\mathbb{E}} \simeq \bigotimes_{j=1}^n \overline{\mathcal{L}}_j.$$

The Griffiths line bundle of the variation is $L_{\mathbb{V}_\mathbb{C}} := \bigotimes_p \det F^p\mathcal{V}$. In weight one, its only contributions come from $F^1\mathcal{V} = \mathbb{E}$, and hence

$$L_{\mathbb{V}_\mathbb{C}} \simeq \det \mathbb{E} = \lambda.$$

It follows that the Deligne–Schmid extension of the Griffiths line bundle is

$$\overline{L}_{\mathbb{V}_\mathbb{C}} := \bigotimes_p \det \overline{F}^p \simeq \det \overline{F}^1 = \overline{\lambda}. \tag{2.2}$$

This is the Griffiths parabolic line bundle whose parabolic weights are all zero in our setting. This line bundle is functorial under pullback. More precisely, if $\overline{Y}$ is smooth, E is a simple normal-crossings divisor, and $f : \overline{Y} \to \overline{X}$ satisfies $f^{-1}(D) \subseteq E$, then, writing $f^\circ : Y := \overline{Y} \setminus E \to X$,

$$\overline{L}_{(f^\circ)^*\mathbb{V}_\mathbb{C}} \simeq f^*\overline{\lambda}.$$

This follows from compatibility of the extended Hodge filtration with pullback; see [Bru20a, §2.2].

Now we identify a multiple of this line bundle with the log-canonical bundle of the toroidal compactification. We use a similar automorphy-factor approach as in [HS02, Section II.1]. By (2.1), $\mathcal{L}_j^{\otimes 2}$ has automorphy factor $j_j(\gamma, z)^2$ and $d(\gamma_j z_j) = j_j(\gamma, z)^{-2} dz_j$. Hence the map on $\mathbb{H}^n$ defined by $s_j(z)^{\otimes 2} \mapsto dz_j$ is Γ-equivariant and descends to an isomorphism

$$\bigoplus_{j=1}^n \mathcal{L}_j^{\otimes 2} \xrightarrow{\sim} \Omega^1_X.$$

We extend this isomorphism across the boundary. Fix a cusp c_i and let $\zeta = (\zeta_1, \dots, \zeta_n)$ be the coordinate around the cusp, so that the translation lattice Λ_i acts by $\zeta \mapsto \zeta + \tau$ for

$\tau \in \Lambda_i$. We use the same notation (e_j, f_j) for the corresponding conjugated flat basis; then $s_j(\zeta) = e_j - \zeta_j f_j$. For $\tau \in \Lambda_i$, one has

$$\rho(\tau)e_j = e_j - \sigma_j(\tau)f_j, \qquad \rho(\tau)f_j = f_j,$$

and hence $s_j(\zeta + \tau) = \rho(\tau)s_j(\zeta)$. Let $\chi_1, \dots, \chi_n$ be a basis of $\Lambda_i^\vee$ and let $q_k := q^{\chi_k}$, $k = 1, \dots, n$, be the toroidal coordinates introduced in Subsection 1.2. Writing $\chi_k(\zeta) = \sum_j a_{kj}\zeta_j$ and $A = (a_{kj})$, we have

$$\begin{pmatrix} dq_1/q_1 \\ \vdots \\ dq_n/q_n \end{pmatrix} = 2\pi\sqrt{-1}\, A \begin{pmatrix} d\zeta_1 \\ \vdots \\ d\zeta_n \end{pmatrix}.$$

Since $\chi_1, \dots, \chi_n$ form a basis of $\Lambda_i^\vee$, the matrix A is invertible. Moreover, since e_j and f_j are flat,

$$\nabla s_j = -d\zeta_j \otimes f_j = -\frac{1}{2\pi\sqrt{-1}} \sum_k (A^{-1})_{jk} \frac{dq_k}{q_k} \otimes f_j, \qquad \nabla f_j = 0.$$

Therefore the bundle generated by s_j and f_j has a logarithmic connection, whose residue along $q_k = 0$ sends

$$s_j \longmapsto -\frac{1}{2\pi\sqrt{-1}} (A^{-1})_{jk} f_j, \qquad f_j \longmapsto 0.$$

Its residues are therefore nilpotent, so this bundle is the canonical Deligne extension and s_j is a local frame of $\overline{\mathcal{L}}_j$. Since A is invertible, the forms $d\zeta_j$ and dq_k/q_k are related by an invertible constant change of basis. Consequently, the map $s_j(\zeta)^{\otimes 2} \mapsto d\zeta_j$ extends without zeros or poles to an isomorphism on the toric cover. This extension is Γ_i/Λ_i-equivariant and hence descends to a neighborhood of the boundary in $\overline{X}$ giving

$$\Omega^1_{\overline{X}}(\log D) \simeq \bigoplus_{j=1}^n \overline{\mathcal{L}}_j^{\otimes 2};$$

see [DT04a, Theorem 5.2(iii)]. Taking determinant gives $K_{\overline{X}}(D) \simeq \bigotimes_{j=1}^n \overline{\mathcal{L}}_j^{\otimes 2} \simeq \overline{\lambda}^{\otimes 2} \simeq \overline{L}_{\mathbb{V}_{\mathbb{C}}}^{\otimes 2}$. In $\mathrm{Pic}(\overline{X})$, this means

$$K_{\overline{X}} + D = 2\overline{L}_{\mathbb{V}_{\mathbb{C}}}. \tag{2.3}$$

We now apply Brunebarbe's higher-dimensional Arakelov inequality. Since the standard weight-one variation has rank $2n$ and length 1, the following is a direct consequence of [Bru20a, Theorems 1.6 and 2.3]:

Proposition 2.1. *Let $Z \subset \overline{X}$ be a positive-dimensional irreducible subvariety not contained in D, and let $\nu : \widetilde{Z} \to Z$ be a log resolution of $(Z, D|_Z)$. Set $\widetilde{D} := \big(\nu^*(D|_Z)\big)_{\mathrm{red}}$. Then*

$$K_{\widetilde{Z}} + \widetilde{D} - \frac{1}{n}\nu^* \left((K_{\overline{X}} + D)|_Z \right)$$

is pseudoeffective.

Proof. Let $f : \widetilde{Z} \to \overline{X}$ be the morphism induced by ν, and let $\mathbb{W}$ be the pullback of $\mathbb{V}_{\mathbb{C}}$ to $\widetilde{Z} \setminus \widetilde{D}$. The standard period map on $\mathbb{H}^n$ is immersive, and f is generically immersive since ν is birational and Z is not contained in D. Hence the period map of $\mathbb{W}$ is generically

immersive. Pullback preserves rank and length, so $\mathbb{W}$ has rank $2n$ and length 1. Therefore [Bru20a, Theorems 1.6 and 2.3] implies that

$$K_{\widetilde{Z}} + \widetilde{D} - \frac{2}{n}\overline{L}_{\mathbb{W}}$$

is pseudoeffective. By functoriality of the Griffiths parabolic line bundle and (2.3),

$$\overline{L}_{\mathbb{W}} \simeq f^*\overline{L}_{\mathbb{V}_{\mathbb{C}}} \simeq \frac{1}{2}f^*(K_{\overline{X}} + D) = \frac{1}{2}\nu^*\left((K_{\overline{X}} + D)|_Z\right);$$

see [Bru20a, §§2.2 and 2.4]. Substitution proves the proposition. □

3. **Hyperbolic volume estimates**

Let $\Gamma \subseteq \Gamma(1)$ be a torsion-free subgroup of finite index, and let $\overline{X}$ be a smooth projective toroidal compactification of $X := X_\Gamma$, with boundary divisor $D := \overline{X} \setminus X_\Gamma$. Throughout the paper, a (sub)variety is assumed to be positive-dimensional and irreducible. We prove in Theorem 3.1 that every subvariety of $\overline{X}$ not contained in D meets the thick part of X. We then establish in Proposition 3.4 a uniform lower bound for the injectivity radius on the thick part of $X_1(\eta)$. Therefore, every such subvariety contains a point with large injectivity radius and then we conclude the hyperbolic volume estimates we need in Theorem 3.5.

Theorem 3.1. *Every subvariety $Z \subseteq \overline{X}$ either intersects X_{thick} or is fully contained in the boundary D.*

We need the following two basic lemmas to prove Theorem 3.1. The first shows that cusp stabilizers of Hilbert modular varieties are solvable:

Lemma 3.2. *Let $P_\infty(\mathbb{R})$ be the upper-triangular group defined in (1.1). Let $\Gamma_\infty \subseteq P_\infty(\mathbb{R})$ be a cusp stabilizer of Γ, and let $\rho : \Gamma_\infty \to \mathrm{GL}(V_\mathbb{Q})$ be a representation on a finite-dimensional $\mathbb{Q}$-vector space. Then the algebraic group*

$$\overline{\rho(\Gamma_\infty)}_\mathbb{Q}^{\mathrm{Zar}} \subseteq \mathrm{GL}(V_\mathbb{Q})$$

is solvable.

Proof. The surjective homomorphism

$$P_\infty(\mathbb{R}) \longrightarrow (\mathbb{R}^\times)^n, \qquad p(a,b) \longmapsto a,$$

has abelian kernel $U_\infty(\mathbb{R}) = \{p(1,b) : b \in \mathbb{R}^n\} \simeq (\mathbb{R}^n, +)$. Since its image is also abelian, $[P_\infty(\mathbb{R}), P_\infty(\mathbb{R})] \subseteq U_\infty(\mathbb{R})$, so $P_\infty(\mathbb{R})$ is metabelian. Therefore Γ_∞ and $\rho(\Gamma_\infty)$ are metabelian, and in particular solvable. Since the Zariski closure of a solvable subgroup is solvable, $\overline{\rho(\Gamma_\infty)}_\mathbb{Q}^{\mathrm{Zar}}$ is solvable. □

We recall the following standard fact about linear algebraic groups:

Lemma 3.3. *Let H be a connected linear algebraic group. If H is both solvable and semisimple, then H is trivial.*

Proof. Let $R(H)$ denote the radical of H, i.e. the maximal connected solvable normal algebraic subgroup of H. Since H is connected and solvable, we have $R(H) = H$. On the other hand, since H is semisimple, its radical is trivial: $R(H) = \{1\}$. Therefore $H = \{1\}$. □

Now we can prove Theorem 3.1:

Proof of Theorem 3.1. Suppose that a subvariety $Z \subset \overline{X}$ not contained in D does not meet X_{thick}. Then the dense open subset $Z^\circ := Z \cap X$ is contained in the union of the canonical horoballs. Since Z° is connected, and these horoballs are pairwise disjoint, it is contained in a single canonical horoball H_i around a cusp c_i. After conjugating Γ, we may assume that c_i is represented by ∞.

Let $\nu : \widetilde{Z} \to Z$ be a projective log resolution of $(Z, D_{|Z})$ such that $\widetilde{D} := \big(\nu^{-1}(Z \cap D)\big)_{\mathrm{red}}$ has simple normal crossings, and set $\widetilde{Z}^\circ := \widetilde{Z} \setminus \widetilde{D}$. The induced map $f : \widetilde{Z}^\circ \to X$ factors through H_i. Since U_i is contractible, Γ_∞ acts freely on U_i, and $H_i = \Gamma_\infty \backslash U_i$, we have $\pi_1(H_i) \simeq \Gamma_\infty$. Therefore, after choosing compatible basepoints, f induces the homomorphism

$$u : \pi_1(\widetilde{Z}^\circ) \longrightarrow \Gamma_\infty \subseteq \Gamma, \qquad \Gamma_\infty \subseteq P_\infty(\mathbb{R}).$$

The monodromy of the pullback of the standard variation is

$$\rho \circ u : \pi_1(\widetilde{Z}^\circ) \longrightarrow \mathrm{GL}(V_{\mathbb{Q}}).$$

Consider the connected algebraic monodromy group

$$\mathbf{H}_Z := \left(\overline{(\rho \circ u)\big(\pi_1(\widetilde{Z}^\circ)\big)}_{\mathbb{Q}}^{\mathrm{Zar}} \right)^\circ \subseteq \mathrm{GL}(V_{\mathbb{Q}}).$$

Since $(\rho \circ u)\big(\pi_1(\widetilde{Z}^\circ)\big) \subseteq \rho(\Gamma_\infty)$, we have $\mathbf{H}_Z \subseteq \overline{\rho(\Gamma_\infty)}_{\mathbb{Q}}^{\mathrm{Zar}}$. By Lemma 3.2, $\overline{\rho(\Gamma_\infty)}_{\mathbb{Q}}^{\mathrm{Zar}}$ is solvable, so $\mathbf{H}_Z$ is solvable.

On the other hand, the pullback of the standard variation to $\widetilde{Z}^\circ$ is a polarized pure $\mathbb{Q}$-variation of Hodge structure. Therefore, André–Deligne semisimplicity [And92, Corollary 1] implies that the group $\mathbf{H}_Z$ is semisimple. It follows from Lemma 3.3 that $\mathbf{H}_Z = \{1\}$. Therefore, $(\rho \circ u)\big(\pi_1(\widetilde{Z}^\circ)\big)$ is finite. Since ρ is faithful and Γ is torsion-free, we obtain

$$u\big(\pi_1(\widetilde{Z}^\circ)\big) = \{1\}.$$

Consequently, f lifts to a holomorphic map $\widetilde{f} : \widetilde{Z}^\circ \longrightarrow \mathbb{H}^n$. Identifying $\mathbb{H}^n$ with the bounded polydisc Δ^n, the coordinate functions of $\widetilde{f}$ are bounded and therefore extend holomorphically across $\widetilde{D}$. The resulting holomorphic functions on the compact connected variety $\widetilde{Z}$ are constant. Hence $\widetilde{f}$, and therefore f, is constant. This contradicts $\dim Z > 0$. □

We denote by $X_{1,\mathrm{thick}}(\eta)$ the thick part of $X_1(\eta)$ defined in Definition 1.5.

Proposition 3.4. *For every $x \in X_{1,\mathrm{thick}}(\eta)$, the following hold:*

(i) *If $|\operatorname{Nm}(\eta)| > 4^n$, then* $\operatorname{inj}_{X_1(\eta)}(x) \geq \dfrac{1}{2n} \log |\operatorname{Nm}(\eta)| - \dfrac{1}{2}$.

(ii) *If $|\operatorname{Nm}(\eta)| \geq 5^n$, then* $\operatorname{inj}_{X_1(\eta)}(x) \geq \operatorname{arsinh}\left(\dfrac{1}{2} |\operatorname{Nm}(\eta)|^{1/(2n)} \right) \geq \dfrac{1}{2n} \log |\operatorname{Nm}(\eta)|$.

Proof. Under either hypothesis, Lemma 1.2 implies that $\Gamma_1(\eta)$ is torsion-free and every parabolic element is unipotent.

Let $z \in \mathbb{H}^n$ be a lift of x, and let $1 \neq \gamma \in \Gamma_1(\eta)$. If γ is semisimple, then [BT18a, Corollary 3.10] gives

$$\frac{1}{2} d_{\mathbb{H}^n}(z, \gamma z) \geq \frac{1}{n} \log |\operatorname{Nm}(\eta)| - 1,$$

and the desired bound follows for this case. Suppose now that γ is parabolic and stabilizes the cusp c_i. In coordinates centered at c_i, write $\gamma z = z + \lambda$ with $0 \neq \lambda \in \Lambda_i$. Setting $y_j = \mathrm{Im}(z_j)$, we have

$$d_{\mathbb{H}}(z_j, z_j + \lambda_j) = 2\sinh^{-1}\left(\frac{|\lambda_j|}{2y_j}\right) \geq 2\log\left(\frac{|\lambda_j|}{y_j}\right).$$

Since the Kobayashi distance on $\mathbb{H}^n$ is the maximum of the coordinatewise distances,

$$\begin{aligned} d_{\mathbb{H}^n}(z, z+\lambda) &\geq 2\max_j \log\left(\frac{|\lambda_j|}{y_j}\right) \\ &\geq \frac{2}{n}\sum_{j=1}^{n}\log\left(\frac{|\lambda_j|}{y_j}\right) = \frac{2}{n}\log\left(\frac{|\,\mathrm{Nm}_i(\lambda)|}{N_i(z)}\right). \end{aligned}$$

By the normalization (1.2), $|\,\mathrm{Nm}_i(\lambda)| \geq 1$. Moreover, since $x \in X_{1,\mathrm{thick}}(\eta)$, $N_i(z)^{-1} \geq d$, where d is the uniform depth of the cusps. Hence [BT18a, Corollary 3.5] yields

$$d_{\mathbb{H}^n}(z, \gamma z) \geq \frac{2}{n}\log d \geq \frac{1}{n}\log|\,\mathrm{Nm}(\eta)|,$$

and the claimed bound for (i) follows.

Suppose now that $|\,\mathrm{Nm}(\eta)| \geq 5^n$. For semisimple γ, the nonzero element $\mathrm{tr}(\gamma) - 2 \in \eta$ has an embedding σ_j satisfying $\big|\sigma_j\big(\mathrm{tr}(\gamma)\big) - 2\big| \geq |\,\mathrm{Nm}(\eta)|^{1/n}$. Therefore

$$\frac{1}{2}d_{\mathbb{H}^n}(z, \gamma z) \geq \mathrm{arcosh}\left(\frac{1}{2}|\,\mathrm{Nm}(\eta)|^{1/n} - 1\right) \geq \sinh^{-1}\left(\frac{1}{2}|\,\mathrm{Nm}(\eta)|^{1/(2n)}\right),$$

where the last inequality follows from $|\,\mathrm{Nm}(\eta)|^{1/n} \geq 5$. If γ is parabolic, set $u_j = |\lambda_j|/y_j$. Then

$$\prod_{j=1}^{n} u_j = \frac{|\,\mathrm{Nm}_i(\lambda)|}{N_i(z)} \geq d \geq |\,\mathrm{Nm}(\eta)|^{1/2},$$

and hence

$$\frac{1}{2}d_{\mathbb{H}^n}(z, \gamma z) = \max_j \mathrm{arsinh}\left(\frac{u_j}{2}\right) \geq \mathrm{arsinh}\left(\frac{1}{2}\left(\prod_{j=1}^{n} u_j\right)^{1/n}\right) \geq \mathrm{arsinh}\left(\frac{1}{2}|\,\mathrm{Nm}(\eta)|^{1/(2n)}\right).$$

The remaining inequality follows from $\mathrm{arsinh}(t/2) \geq \log t$ for $t > 0$. This proves (ii). □

Now we can conclude a uniform bound on the degrees of all subvarieties with respect to the log-canonical bundle:

Theorem 3.5. *Suppose that $|\,\mathrm{Nm}(\eta)| > 4^n$. Let $\overline{X_1}(\eta)$ be a smooth toroidal compactification of $X_1(\eta)$ with boundary divisor D and let Z be an irreducible subvariety of $\overline{X_1}(\eta)$ which is not contained in D. Then,*

$$(K_{\overline{X_1}(\eta)} + D)^m \cdot Z \geq 2^m \sinh^{2m}\big(\frac{1}{4n}\log|\,\mathrm{Nm}(\eta)| - \frac{1}{4}\big),$$

where m is the dimension of Z. Moreover, if $|\,\mathrm{Nm}(\eta)| \geq 5^n$, then

$$(K_{\overline{X_1}(\eta)} + D)^m \cdot Z \geq 2^m \sinh^{2m}\left(\frac{1}{2}\,\mathrm{arsinh}\left(\frac{1}{2}|\,\mathrm{Nm}(\eta)|^{1/(2n)}\right)\right).$$

Proof. By Theorem 3.1, the variety Z intersects $X_{1,\mathrm{thick}}(\eta)$ and therefore by Proposition 3.4 the subvariety Z contains a point with injectivity radius at least $\frac{1}{2n}\log|\operatorname{Nm}(\eta)|-\frac{1}{2}$ on $X_1(\eta)$. Hence, Hwang-To's inequality (1.5), together with (1.4), provides us a uniform bound for the intersection of Z with the log-canonical bundle:

$$(K_{\overline{X_1(\eta)}}+D)^m\cdot Z=\frac{m!}{(2\pi)^m}\operatorname{vol}_{X_1(\eta)}(Z)\geq 2^m\sinh^{2m}\Big(\frac{1}{4n}\log|\operatorname{Nm}(\eta)|-\frac{1}{4}\Big).$$

If $|\operatorname{Nm}(\eta)|\geq 5^n$, the same argument using Proposition 3.4(ii) gives the second part. □

Remark 3.6. Note that the right-hand side of the inequalities in Theorem 3.5 depends only on the norm of the ideal η and the degree of the field of multiplication over $\mathbb{Q}$ and therefore it is independent of the choice of field or the compactification.

4. Canonical Volume

The volume of a line bundle L on an m-dimensional projective variety V is defined to be the non-negative real number

$$\operatorname{vol}_V(L):=\limsup_{b\to\infty}\frac{h^0(V,bL)}{b^m/m!},$$

which measures the positivity of L from the point of view of birational geometry. If L is a nef line bundle on V, then $\operatorname{vol}_V(L)=L^m$. Also, L is big if and only if $\operatorname{vol}_V(L)>0$.

The canonical volume of a quasi-projective Z is the volume of the canonical bundle of any smooth projective variety birational to Z. More explicitly, if $\overline{Z}$ is a smooth projective variety birational to Z, then

$$\widetilde{\operatorname{vol}}_Z:=\operatorname{vol}_{\overline{Z}}(K_{\overline{Z}}).$$

is the canonical volume of Z. This quantity is independent of the choice of the smooth projective model $\overline{Z}$. In particular, when Z is an integral curve, then $\widetilde{\operatorname{vol}}_Z=\max\{2g-2,0\}$, where g is the geometric genus of Z.

Remark 4.1. Let $f\colon Z\dashrightarrow Z'$ be a dominant generically finite rational map of degree d between integral quasi-projective varieties. After passing to smooth projective models, we obtain a generically finite morphism $\overline{f}\colon\overline{Z}\to\overline{Z'}$. The ramification formula gives $K_{\overline{Z}}=\overline{f}^*K_{\overline{Z'}}+R$ for some effective divisor R. Hence

$$\widetilde{\operatorname{vol}}_Z=\operatorname{vol}(K_{\overline{Z}})\geq\operatorname{vol}(\overline{f}^*K_{\overline{Z'}})=d\,\operatorname{vol}(K_{\overline{Z'}})=d\,\widetilde{\operatorname{vol}}_{Z'}.$$

The bridge between degrees with respect to the log-canonical bundle and canonical volumes is provided by the following proposition, which follows from the Arakelov inequality proved by Brunebarbe [Bru20a, Theorem 1.6]. Brunebarbe's result applies in a more general setting, where Griffiths transversality is a non-trivial condition. In the setting of Shimura varieties, however, this condition is automatically satisfied, and his theorem gives the following Proposition:

Proposition 4.2. *Suppose that $|\operatorname{Nm}(\eta)|>4^n$. Then, for every m-dimensional subvariety $Z\subset\overline{X}_1(\eta)$ not contained in D, we have*

$$\widetilde{\operatorname{vol}}_Z\geq\frac{1}{n^m}\operatorname{vol}_Z\Big((K_{\overline{X}_1(\eta)}-(n-1)D)\Big|_Z\Big).$$

Proof. Let $\nu : \widetilde{Z} \to Z$ be a log resolution of $(Z, D|_Z)$ and set $\widetilde{D} = (\nu^*(D|_Z))_{\mathrm{red}}$. By Proposition 2.1, we know that $K_{\widetilde{Z}} + \widetilde{D} - \frac{1}{n}\nu^*\left((K_{\overline{X}_1(\eta)} + D)\Big|_Z\right)$ is pseudoeffective. Since $\nu^*(D|_Z) - \widetilde{D}$ is effective, it follows that $K_{\widetilde{Z}} - \frac{1}{n}\nu^*\left((K_{\overline{X}_1(\eta)} - (n-1)D)\Big|_Z\right)$ is pseudoeffective. Since the volume does not decrease in pseudoeffective direction we get

$$\widetilde{\mathrm{vol}}_Z = \mathrm{vol}_{\widetilde{Z}}(K_{\widetilde{Z}}) \geq \mathrm{vol}_Z\left(\frac{1}{n}\left(K_{\overline{X}_1(\eta)} - (n-1)D\right)\Big|_Z\right).$$

□

The stable base locus of a line bundle L is defined as

$$\mathbf{B}(L) := \bigcap_{m \geq 1} \mathrm{Bs}(L^{\otimes m}).$$

Given an ample line bundle A, the augmented base locus of L is

$$\mathbf{B}_+(L) := \bigcap_{\varepsilon > 0} \mathbf{B}(L - \varepsilon A),$$

where ε ranges over sufficiently small positive rational numbers. This definition is independent of the choice of A.

Definition 4.3. A line bundle L is called *ample modulo* a closed subset E if $\mathbf{B}_+(L) \subset E$.

Lemma 4.4. *Let Y be a projective variety, let $E \subset Y$ be a closed subset, and let L be a line bundle on Y. Suppose that L is ample modulo E. Then for every irreducible subvariety $Z \subset Y$ with $Z \not\subset E$, the restricted line bundle $L|_Z$ is big.*

Proof. The proof is similar to the proof of [Mem22, Lemma 4.7]. Since $Z \not\subset E$ and $\mathbf{B}_+(L) \subset E$, we have $Z \not\subset \mathbf{B}_+(L)$. Choose an ample line bundle A on Y. For $m \gg 0$,

$$\mathbf{B}_+(L) = \mathrm{Bs}(mL - A).$$

Thus Z is not contained in $\mathrm{Bs}(mL - A)$, so there exists a section of $mL - A$ which does not vanish identically on Z. Hence $(mL - A)|_Z$ has a nonzero section. Therefore

$$mL|_Z \cong A|_Z + \Sigma$$

for some effective divisor Σ on Z. Since $A|_Z$ is ample, it is big. Adding an effective divisor preserves bigness, so $mL|_Z$ is big. Hence $L|_Z$ is big. □

We will use the following positivity result of Bakker-Tsimerman:

Proposition 4.5. *([BT18a, Theorem E]) Suppose that $|\mathrm{Nm}(\eta)| > 4^n$. If $\lambda > 0$ and $|\mathrm{Nm}(\eta)| > \left(\frac{2\pi\lambda}{n}\right)^{2n}$, then the divisor $K_{\overline{X_1(\eta)}} + (1-\lambda)D$ is ample modulo D.*

Lemma 4.6. *Suppose $c > n/\pi$ and set $N(n,c) := 1 + \lfloor (2\pi c/n)^{2n} \rfloor$. If $|\mathrm{Nm}(\eta)| \geq N(n,c)$, then for every subvariety $Z \subset \overline{X_1}(\eta)$ intersecting with $X_1(\eta)$, the divisor $\left(K_{\overline{X_1(\eta)}} + (1-c)D\right)|_Z$ is big.*

Proof. By Proposition 4.5, $K_{\overline{X_1(\eta)}} + (1-c)D$ is ample modulo D. Since $Z \not\subset D$, the conclusion follows from Lemma 4.4. □

Let Z be an integral quasi-projective variety over $k = \mathbb{C}$, and set $K = k(Z)$. The generic fiber A_z of an abelian scheme $A \to Z$ is an abelian variety over K. Conversely, every abelian variety over K extends to an abelian scheme over some nonempty open subset $U \subset Z$. We denote by $A(Z) := A_z(K)$ the Mordell–Weil group of rational sections of $A \to Z$, and by $A(Z)_{\mathrm{tor}}$ its torsion subgroup.

After replacing Z by a nonempty open subset, fix the polarization and polarized $\mathcal{O}_F$-module type M as in Remark 1.1; thus $X(1)$ below denotes the corresponding Hilbert modular variety.

Fix a totally real field F. The Hilbert modular variety $X(1)$ parametrizes abelian varieties with real multiplication by $\mathcal{O}_F$; see Section 1.1. Accordingly, a family $A \to Z$ of such abelian varieties determines a moduli map $\mu_A \colon Z \longrightarrow X(1)$. We say that $A \to Z$ has *no isotrivial directions* if μ_A is generically finite onto its image. When Z is a curve, this is equivalent to non-isotriviality, while in higher dimensions it excludes directions in the base along which the family is constant. For any ideal $\eta \subset \mathcal{O}_F$, the torsion level cover $X_1(\eta)$ parametrizes such abelian varieties together with a point whose annihilator is η. A torsion section $x \in A(Z)$ with $\mathrm{Ann}_{\mathcal{O}_F}(x) = \eta$ induces a lift $\mu_{A,x} \colon Z \longrightarrow X_1(\eta)$ of μ_A. If $A \to Z$ has no isotrivial directions, then $\mu_{A,x}$ is also generically finite onto its image. Therefore, the canonical volume of Z is greater than or equal to the canonical volume of its image in $X_1(\eta)$; see Remark 4.1, and we will still denote the image of Z in $X_1(\eta)$ by Z.

Given our current setup, to prove Theorem A, it is enough to prove the following theorem:

Theorem 4.7. *Given a real number $v \geq 0$ and a positive integer n, there exists an integer $M = M(n, v)$ such that, for every totally real field $F/\mathbb{Q}$ of degree n, every ideal $\eta \subset \mathcal{O}_F$ satisfying $|\operatorname{Nm}(\eta)| \geq M$, and every smooth toroidal compactification $\overline{X_1}(\eta)$ of $X_1(\eta)$, every positive-dimensional subvariety $Z \subset \overline{X_1}(\eta)$ intersecting with $X_1(\eta)$ has canonical volume $> v$. Explicitly, one may take*

$$M(n, v) := \max\left\{1 + \lfloor (4\pi)^{2n} \rfloor, \left\lceil \big(16n \max\{1, v\}\big(1 + n \max\{1, v\}\big)\big)^n \right\rceil\right\}.$$

Remark 4.8. If $\eta = \mathrm{Ann}_{\mathcal{O}_F}(x)$, then the order of x is at most $|\operatorname{Nm}(\eta)|$. Theorem 4.7 shows that every torsion point has order less than $M(n, v)$. Since $M(n, v) \leq \left\lceil (4\pi n \max\{1, v\})^{2n} \right\rceil$, Theorem A follows by taking

$$N(n, v) = \big(\left\lceil (4\pi n \max\{1, v\})^{2n} \right\rceil - 1\big)!.$$

Proof of Theorem 4.7. Let $Z \subset \overline{X_1}(\eta)$ be an arbitrary m-dimensional subvariety intersecting with $X_1(\eta)$. Applying Lemma 4.6 with $c = 2n$ shows that, if $|\operatorname{Nm}(\eta)| \geq N_1 := 1 + \lfloor (4\pi)^{2n} \rfloor$, then the divisor $\big(K_{\overline{X_1}(\eta)} - (2n-1)D\big)|_Z$ is big. Notice that $N_1 > 5^n$, so the second part of Proposition 3.4 applies. We can write

$$K_{\overline{X_1}(\eta)} - (n-1)D = \frac{1}{2}\big(K_{\overline{X_1}(\eta)} - (2n-1)D\big) + \frac{1}{2}\big(K_{\overline{X_1}(\eta)} + D\big).$$

Note that $K_{\overline{X_1}(\eta)} + D$ is nef on $\overline{X_1}(\eta)$ because it is a pull back of an ample bundle from a Baily-Borel compactification; see [Mum77, Proposition 3.4.b]. Therefore, its restriction to Z is nef and, by Theorem 3.5, big. By the log-concavity of the volume function on the big

cone of Z (see, for example, [Laz04, Theorem 11.4.9]), we obtain

$$\begin{aligned}\operatorname{vol}_Z\Big(\big(K_{\overline{X_1(\eta)}}-(n-1)D\big)|_Z\Big) &> \operatorname{vol}_Z\left(\frac{1}{2}\big(K_{\overline{X_1(\eta)}}+D\big)|_Z\right)\\ &= (2)^{-m}(K_{\overline{X_1(\eta)}}+D)^m\cdot Z \qquad \text{(by nefness)}\\ &\geq \sinh^{2m}\left(\frac{1}{2}\operatorname{arsinh}\left(\frac{1}{2}|\operatorname{Nm}(\eta)|^{1/(2n)}\right)\right) \qquad \text{(by Theorem 3.5)}.\end{aligned}$$

Therefore, Proposition 4.2 gives

$$\widetilde{\operatorname{vol}}_Z > n^{-m}\sinh^{2m}\left(\frac{1}{2}\operatorname{arsinh}\left(\frac{1}{2}|\operatorname{Nm}(\eta)|^{1/(2n)}\right)\right).$$

The constant $N_2 := \left\lceil\big(16n\max\{1,v\}\big(1+n\max\{1,v\}\big)\big)^n\right\rceil$ ensures that the right-hand side is at least v, uniformly for $1\leq m\leq n$. Therefore, taking $M=\max\{N_1,N_2\}$ proves the theorem. □

A variety is of general type if and only if its canonical volume is positive. Similar to Theorem 4.7 we get the following:

Corollary 4.9. *Let $F/\mathbb{Q}$ be a totally real field of degree n. Every subvariety of $X_1(\eta)$ is of general type provided that $|\operatorname{Nm}(\eta)|>(2\pi)^{2n}$.*

Proof. Let $Z\subset\overline{X_1}(\eta)$ be an m-dimensional subvariety meeting $X_1(\eta)$. Lemma 4.6, applied with $c=n$, shows that $(K_{\overline{X_1(\eta)}}-(n-1)D)|_Z$ is big. Hence, Proposition 4.2 gives

$$\widetilde{\operatorname{vol}}_Z \geq \frac{1}{n^m}\operatorname{vol}_Z\Big(\big(K_{\overline{X_1(\eta)}}-(n-1)D\big)|_Z\Big)>0.$$

Therefore Z is of general type. □

5. Sparsity of rational points

Let K be a number field. On projective space, we use the absolute multiplicative height

$$\operatorname{H}([x_0:\cdots:x_N]) = \prod_{v\in M_K}\max_i|x_i|_v^{[K_v:\mathbb{Q}_v]/[K:\mathbb{Q}]},$$

where the absolute values extend the standard ones on $\mathbb{Q}$. This is independent of both the choice of homogeneous coordinates and the number field containing them. The problem of bounding K-rational points of bounded height on algebraic varieties is closely related to the determinant method, originating in the work of Bombieri–Pila [BP89] and further developed by Heath-Brown [HB02]. As observed in [ELV23, BM24], uniform lower bounds for the degrees of the subvarieties arising in this process lead to improved bounds for the growth rate of rational points; see also [Chi25] for a mixed-variation setting. In particular, we recall the following theorem of Brunebarbe–Maculan:

Theorem 5.1. *([BM24, Theorem 4.4]) Let E be a subvariety of $\mathbb{P}^N_K$, let $\epsilon>0$ be a real number, let $n\geq 0$ and $e\geq 1$ be integers. Then, there is a real number $C=c(n,e,N,K,E,\epsilon)$ with the following property: For an n-dimensional subvariety Y of $\mathbb{P}^N$ of degree $\leq e$ such*

that each subvariety Z in Y not contained in E has degree $\geq \rho^{\dim(Z)}$ for some integer $\rho \geq 1$, and a real number $B > e^{[K:\mathbb{Q}]\epsilon}$, the following inequality holds:

$$\#\{x \in Y(K) \setminus E \mid \mathrm{H}(x) \leq B\} \leq CB^{(1+\epsilon)[K:\mathbb{Q}]n(n+3)/(2\rho)}.$$

Remark 5.2. There is a typo in the statement of Theorem 5.1 in the original paper. However, from its applications and the surrounding discussion in that paper, it is clear that the exponent of ρ should be the dimension of the subvariety Z, not E.

Let $\overline{Y}$ be a projective variety over K, let $E \subset \overline{Y}$ be a closed subset, and let L be a semiample line bundle which is ample modulo E; see Definition 4.3. We fix once and for all an integer $b \geq 1$ such that bL is base-point free and a basis of its global sections, defining a morphism $\varphi_b : \overline{Y} \longrightarrow \mathbb{P}^N_K$. We define

$$H_L(x) := \mathrm{H}(\varphi_b(x))^{1/b}.$$

The resulting height depends on the choice of b and the basis, but if H_L and H'_L are heights arising from two such choices, then there exists a constant $C_0 \geq 1$ such that

$$C_0^{-1} H_L(x) \leq H'_L(x) \leq C_0 H_L(x)$$

for every $x \in \overline{Y}(K)$. We keep the above choice of H_L fixed throughout.

Lemma 5.3. *Let $n = \dim \overline{Y}$, and suppose that the morphism φ_b identifies $\overline{Y} \setminus E$ with the complement of a closed subset E' in its projective image. Assume that there exists a real number $\rho > 0$ such that $L^m \cdot V \geq \rho^m$ for every subvariety $V \subset \overline{Y}$ not contained in E, where $m = \dim V$. Then, for every $\epsilon > 0$, there exists a constant $C_\epsilon > 0$ such that*

$$\#\{x \in (\overline{Y} \setminus E)(K) : H_L(x) \leq B\} \leq C_\epsilon B^{(1+\epsilon)[K:\mathbb{Q}]n(n+3)/(2\rho)}$$

for every $B \geq 1$.

Proof. Compose φ_b with the q-th Veronese embedding. If Z is a subvariety of the resulting projective model not contained in E', and $V \subset \overline{Y}$ is its strict transform, then the projection formula gives

$$\deg(Z) = (qbL)^m \cdot V \geq (qb\rho)^m,$$

where $m = \dim Z$. Fix $\epsilon > 0$. For q sufficiently large, we may choose an integer $d < qb\rho$ such that $(1 + \epsilon/2)\frac{qb}{d} \leq \frac{1+\epsilon}{\rho}$; for example, one may take $d = \lceil qb\rho \rceil - 1$. Hence $\deg(Z) > d^{\dim Z}$, and Theorem 5.1, applied with $\epsilon/2$, gives

$$\begin{aligned}\#\{x \in (\overline{Y} \setminus E)(K) : H_L(x) \leq B\} &\leq C\big(B^{qb}\big)^{(1+\epsilon/2)[K:\mathbb{Q}]n(n+3)/(2d)} \\ &\leq CB^{(1+\epsilon)[K:\mathbb{Q}]n(n+3)/(2\rho)}.\end{aligned}$$

Here we used that the height associated with the Veronese embedding is H_L^{qb}. Once q is chosen, the projective model is fixed, and hence the resulting constant C_ϵ is independent of B. Enlarging C, if necessary, covers the bounded range of B excluded by Theorem 5.1. □

Arithmetic model. The purpose of this standard discussion is only to place the varieties and line bundles used below over $\mathbb{Q}$, so that the height argument is well-defined. We follow the framework of Dimitrov–Tilouine [DT04b, §1], specialized to our setting and notation. For the levels in Theorem 5.4, choose an integral ideal $\mathfrak{c}$ in the narrow ideal class determined by the fixed polarized type, prime to $\mathrm{Nm}(\eta)\mathcal{O}_F$; this is possible by weak approximation [DT04b, Remark 1.5]. Since η is relatively prime to Δ_F, it is also relatively prime to $\mathrm{Nm}(\mathfrak{c}\mathfrak{d}_F)\mathcal{O}_F$. Moreover, if $\eta \mid (2)$ or $\eta \mid (3)$, then respectively $|\mathrm{Nm}(\eta)| \le 2^n$ or $|\mathrm{Nm}(\eta)| \le 3^n$, contrary to $|\mathrm{Nm}(\eta)| \ge 5^n$. Thus both conditions of Dimitrov–Tilouine's hypothesis (NT) hold. For every $\mathbb{Q}$-algebra R, consider the determinant map

$$\det\colon\ \mathrm{GL}_2(F \otimes_{\mathbb{Q}} R) \longrightarrow (F \otimes_{\mathbb{Q}} R)^{\times}.$$

Define $\mathsf{G}(R) := \{g \in \mathrm{GL}_2(F \otimes_{\mathbb{Q}} R) \mid \det(g) \in R^{\times}\}$, where $R^{\times}$ is embedded diagonally in $(F \otimes_{\mathbb{Q}} R)^{\times}$ by $a \mapsto 1 \otimes a$. This group is represented by the fiber product

$$\mathsf{G} = \mathrm{Res}_{F/\mathbb{Q}}\,\mathrm{GL}_2 \times_{\mathrm{Res}_{F/\mathbb{Q}}\,\mathbb{G}_m} \mathbb{G}_m,$$

taken with respect to the determinant and the diagonal inclusion. The determinant yields a split exact sequence

$$1 \longrightarrow \mathrm{Res}_{F/\mathbb{Q}}\,\mathrm{SL}_2 \longrightarrow \mathsf{G} \xrightarrow{\det} \mathbb{G}_m \longrightarrow 1,$$

with section $a \mapsto \mathrm{diag}(1, a)$. Since the quotient $\mathbb{G}_m$ is commutative and $\mathrm{Res}_{F/\mathbb{Q}}\,\mathrm{SL}_2$ is perfect, we have $\mathsf{G}^{\mathrm{der}} = \mathrm{Res}_{F/\mathbb{Q}}\,\mathrm{SL}_2$.

Let $(\mathsf{G}, \mathcal{H})$ be the associated Hilbert Shimura datum, and let $\mathcal{H}^+ = \mathbb{H}^n$ be its distinguished connected component. Let M be the rank-two projective $\mathcal{O}_F$-module fixed in Section 1.1, and set $\widehat{M} := M \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}}$, and $e_1 := (1, 0)$. Let

$$K_\eta := \left\{ g \in \mathsf{G}(\mathbb{A}_f) \;\middle|\; g\widehat{M} = \widehat{M}, \quad g(e_1) \equiv e_1 \pmod{\eta\widehat{M}} \right\}.$$

Writing $\mathsf{G}(\mathbb{Q})^+ := \{g \in \mathsf{G}(\mathbb{Q}) : \det(g) > 0\}$, we have

$$\mathsf{G}(\mathbb{Q})^+ \cap K_\eta = \Gamma_1(\eta), \qquad \det\big(\mathsf{G}(\mathbb{Q})^+\big) = \mathbb{Q}^{\times}_{>0}, \qquad \det(K_\eta) = \widehat{\mathbb{Z}}^{\times}.$$

Strong approximation for $\mathsf{G}^{\mathrm{der}}$, together with the exact sequence above, gives

$$\mathsf{G}(\mathbb{Q})^+ \backslash \mathsf{G}(\mathbb{A}_f) / K_\eta \overset{\det}{\simeq} \mathbb{Q}^{\times}_{>0} \backslash \mathbb{A}_f^{\times} / \widehat{\mathbb{Z}}^{\times} = \{1\};$$

see [Mil05, Theorem 4.16]. It follows from the complex uniformization that

$$\mathrm{Sh}_{K_\eta}(\mathsf{G}, \mathcal{H})(\mathbb{C}) \simeq \big(\mathsf{G}(\mathbb{Q})^+ \cap K_\eta\big) \backslash \mathcal{H}^+ = \Gamma_1(\eta) \backslash \mathbb{H}^n = X_1(\eta).$$

The defining cocharacter has identical components at all embeddings $F \hookrightarrow \mathbb{R}$, so its conjugacy class is Galois invariant and the reflex field of $(\mathsf{G}, \mathcal{H})$ is $\mathbb{Q}$. Canonical-model theory therefore endows $X_1(\eta)$ with a model over $\mathbb{Q}$; see [Mil05, §§12–14]. Moreover, by [Pin90, Theorem 12.3], its Baily–Borel compactification $X_1(\eta)^*$ is a normal projective variety over $\mathbb{Q}$. We use these models throughout this section.

Counting result. Let $\overline{X}_1(\eta)$ be a smooth toroidal compactification fixed in Section 1.1, with boundary divisor D, and consider the log-canonical bundle on $\overline{X}_1(\eta)$:

$$L := K_{\overline{X}_1(\eta)} + D.$$

Let $\pi\colon \overline{X}_1(\eta) \longrightarrow X_1(\eta)^*$ be the Baily–Borel contraction. By [DT04b, Theorem 5.2], there exist $b \geq 1$ and a very ample line bundle A on $X_1(\eta)^*$, defined over $\mathbb{Q}$, such that $\pi^*A \simeq bL$. Fix a closed immersion $\iota\colon X_1(\eta)^* \hookrightarrow \mathbb{P}^N_{\mathbb{Q}}$ defined by A, and set

$$H_A(x) := H\big(\iota(x)\big), \qquad H_L(x) := H_A(x)^{1/b}$$

for $x \in X_1(\eta)(\overline{\mathbb{Q}})$. We keep these choices fixed. Since A is very ample, H_L has the Northcott property on $X_1(\eta)(K)$ for every number field K. For further properties of this height see [Mem25, §8].

Theorem 5.4. *Assume that* $|\operatorname{Nm}(\eta)| \geq 5^n$ *and that* η *is relatively prime to the discriminant* Δ_F*. For every number field* K*, set*

$$N_{\eta,K}(B) = \#\{x \in X_1(\eta)(K) : H_L(x) \leq B\}.$$

Then

$$\limsup_{B\to\infty} \frac{\log \max\{1, N_{\eta,K}(B)\}}{\log B} \leq \delta_{\eta,K,n},$$

where

$$\delta_{\eta,K,n} = \frac{\sqrt{5}[K:\mathbb{Q}]n(n+3)}{|\operatorname{Nm}(\eta)|^{1/(2n)}}.$$

Remark 5.5 (**Sparsity of rational points**)**.** In Theorem 5.4, $\delta_{\eta,K,n} \to 0$ uniformly when n and $[K:\mathbb{Q}]$ are bounded and $|\operatorname{Nm}(\eta)| \to \infty$ through ideals prime to Δ_F. Therefore, as we go higher in the tower $X_1(\eta)$, its K-rational points become sparser.

Proof. Set $s_\eta := \sinh^2\left(\frac{1}{2}\operatorname{arsinh}\left(\frac{1}{2}|\operatorname{Nm}(\eta)|^{1/(2n)}\right)\right)$ and $E := X_1(\eta)^* \setminus X_1(\eta)$. Let $Z \subset X_1(\eta)^*$ be an m-dimensional subvariety not contained in E, with $m > 0$, and let $\widetilde{Z} \subset \overline{X}_1(\eta)$ be the strict transform of $Z_{\mathbb{C}}$. Since π is an isomorphism over $X_1(\eta)$, the projection formula and Theorem 3.5 give

$$\deg_A(Z) = \big(\pi^*A\big)^m \cdot \widetilde{Z} = b^m L^m \cdot \widetilde{Z} \geq (2bs_\eta)^m.$$

Applying Lemma 5.3 to $X_1(\eta)^*$, E, and A, with $\rho = 2bs_\eta$, and using $H_L = H_A^{1/b}$, we obtain, for every $\epsilon > 0$,

$$N_{\eta,K}(B) \leq C_\epsilon B^{(1+\epsilon)[K:\mathbb{Q}]n(n+3)/(4s_\eta)}.$$

It follows that

$$\limsup_{B\to\infty} \frac{\log \max\{1, N_{\eta,K}(B)\}}{\log B} \leq \frac{[K:\mathbb{Q}]n(n+3)}{4s_\eta}.$$

Finally, $4s_\eta \geq \frac{1}{\sqrt{5}}|\operatorname{Nm}(\eta)|^{1/(2n)}$, since $|\operatorname{Nm}(\eta)|^{1/(2n)} \geq \sqrt{5}$, which gives the stated bound. □

An important arithmetic ingredient in both [ELV23, BM24] is an integral Chevalley–Weil argument. Namely, an integral point determines an unramified torsor arising from the fiber of a finite étale cover; Hermite–Minkowski implies that only finitely many such torsors occur, and twisting by these torsors allows every integral point to lift to a rational point on one of finitely many twists. We conclude this section with a remark explaining why the same

argument does not directly apply to rational points on quasi-projective varieties, and hence why we adopt an intrinsic approach in this paper.

Remark 5.6 (**Integral versus Rational Chevalley–Weil**)**.** Let X be a quasi-projective variety over a number field K, and let $f\colon Y \to X$ be a finite étale Galois cover with Galois group $G := \mathrm{Aut}_X(Y)$. By spreading out, there exist a finite set S of places of K and $\mathcal{O}_{K,S}$-models $\mathcal{X}$ and $\mathcal{Y}$ of X and Y such that f extends to a finite étale G-torsor $\mathcal{Y} \to \mathcal{X}$, where G is regarded as a constant finite group scheme. For every $x \in \mathcal{X}(\mathcal{O}_{K,S})$, the fiber

$$\mathcal{Y}_x := \mathcal{Y} \times_{\mathcal{X}} \operatorname{Spec} \mathcal{O}_{K,S}$$

defines a class in $H^1(\mathcal{O}_{K,S}, G)$. By Hermite–Minkowski, this set is finite; see [BM24, Proposition 4.6] or [ELV23, Proof of Claim 2]. Twisting $Y \to X$ by representatives of these finitely many classes therefore produces finitely many covers $Y_i \to X$ such that every S-integral point of X lifts, in fact integrally, to some Y_i. This is the form of the Chevalley–Weil argument used in [ELV23, §3] and [BM24, Proposition 4.12].

For rational points on a nonproper variety, however, the corresponding torsors need not be unramified outside any fixed finite set of places. For example, under the finite étale morphism

$$\mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad t \longmapsto t^2,$$

the fiber above $a \in K^\times$ is the torsor defined by $t^2 = a$. Already for $K = \mathbb{Q}$ and $a = p$, where p is an odd prime, the field $\mathbb{Q}(\sqrt{p})$ is ramified at p. Therefore, as p varies, the resulting torsor classes are ramified at infinitely many places and are pairwise distinct. By contrast, if $a \in \mathcal{O}_{K,S}^\times$ and S contains every finite place of K above 2, then

$$\operatorname{Spec} \mathcal{O}_{K,S}[t]/(t^2 - a) \longrightarrow \operatorname{Spec} \mathcal{O}_{K,S}$$

is finite étale, since its discriminant $4a$ is a unit. Hence the associated torsor is unramified outside S.

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